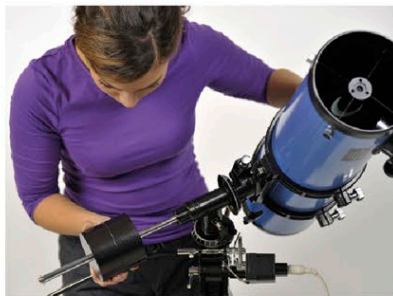




**11 FIT THE FINDER AND EYEPIECE**  
Attach the lowest-magnification eyepiece (the one with the longest focal length) and fit the finder. Align your finder with the main instrument; ideally, do this during the daytime (see p.84).



**12 BALANCE RIGHT ASCENSION (RA)**  
Supporting the telescope with one hand, loosen the clamp on the RA axis and disengage the motor drive if necessary. Adjust the position of the counterweight on its shaft until it balances the telescope and the axis turns freely.



**13 BALANCE DECLINATION (DEC)**  
Position the mount so that the telescope is out to one side and loosen the declination axis clamp. Support the telescope. Slacken the cradle screws enough to be able to slide the tube back and forth until it is balanced, then tighten them.



**14 POWER UP**  
Connect the motor to the power supply, then fine-tune the polar alignment (see below).

## ALIGNING TO THE POLE

To set up an equatorial mount, you need to direct the RA axis, or polar axis, at the celestial pole. How you do this depends on your instrument. With simpler instruments, use the latitude scale usually provided on the mount (see below). On advanced instruments, sight the known position of the pole in the sky (see right). The pole is due south or north, depending on your hemisphere, and at the same angle to the horizon as your geographical latitude. For most observing, approximate alignment by eye is good enough to allow objects to be tracked for many minutes.



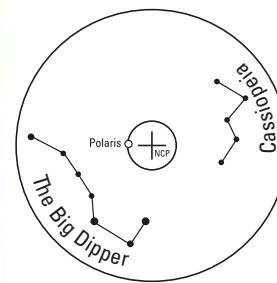
### ALIGNING SIMPLE MOUNTS

Point the polar axis to the north or south depending on hemisphere. Turn the adjuster until the angle on the scale is at your latitude.



### ALIGNING ADVANCED MOUNTS

Looking through a Northern Hemisphere polarscope (left), you will see a reticule engraved with several constellations and a circle offset from a crosshair (below). Turn the reticule until the constellations match their positions in the sky, then adjust the whole mount so that Polaris sits in the small circle.



**1 MOUNT THE TELESCOPE**  
Use a level to check that the tripod is level. Gently lower the mount and telescope onto the tripod head and secure it in position.



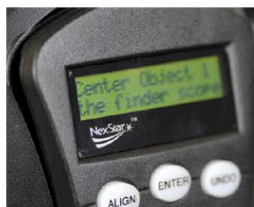
**2 SET POSITION**  
Move the telescope into its start position if required. For a fork-mounted instrument (shown), this may just mean aligning two arrows; however, an equatorial mount will need polar alignment (see above right).



**3 PREPARE THE TELESCOPE**  
Connect the mount to the power pack and switch on the mount. Remove the lens cover.



**4 ENTER START DATA**  
Enter the date, time, and location into the handset as prompted. On some go-to telescopes, you select your location from a menu.



**5 ALIGN THE TELESCOPE**  
Alignment methods vary between models, but typically, the instrument will choose a bright star and move automatically to where it thinks the star should be. Alternatively, you can choose the first star from the menu.



**6 ADJUST ALIGNMENT**  
The first star should be visible in the finder. Center this star using the directional buttons on the handset, then look through the eyepiece to refine its position. Repeat steps 5 and 6 to align two or three more stars as required, then alignment is complete.



**7 SET THE DESTINATION**  
The go-to telescope is now ready. To explore the sky, find the name of the object you want to observe (such as Jupiter) in the handset's menu and press "go-to" or "enter." The telescope will then move to center your chosen object in the eyepiece.

## SETTING UP GO-TO TELESCOPES

Every go-to telescope has a virtual map of the sky in its memory so that once it knows its precise location, the time, and the direction it is pointing to, it can find any celestial object. Encoders on each axis count the number of motor rotations the instrument makes as it "drives" from one object to another. With some simple go-to telescopes (see below), time and location must be input before the instrument is set in its start position, such as leveled and pointed north or south. It then has to be pointed at three bright alignment stars. Depending on the model, these may be chosen from the telescope's catalog or simply any three bright stars or planets. Advanced models are fitted with a GPS (Global Positioning System) receiver that automatically sets the time and location, as well as cameras that locate known bright stars. Whichever type of go-to telescope you have, it is essential that you align the finder with your main instrument before you begin (see p.84).